



DEPARTMENT OF MATHEMATICS - 12 September 2019

MATHEMATICAL MODELLING FOR INDUSTRY

ASSIGNMENT 1:

IMAGE DENOISING

Instructor: Prof. S Sundar

By:
Karl Darryl Lewis
MA18C014

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1 Introduction

This assignment compares various types of diffusion filters. The techniques explored here are as follows: Linear diffusion filtering, Perona Malik model, Edge Enhancing Diffusion(EED) and Coherence Enhancing Diffusion.

The process of image enhancement broadly involves suppressing the noise while preserving the edges. Hence, basically it is a low pass filtering process. The physical phenomenon that imitates our requirement is that of diffusion. Cleaning of images uniformly over all directions is not very effective in image restoration. This is because the filter doesn't take into consideration the edges that need to be preserved.

2 Image denoising techniques and MATLAB Simulation results

2.1 Linear diffusion

Linear diffusion process can be classified into Linear Isotropic Diffusion and Linear Anisotropic diffusion.

Linear diffusion process in images is equivalent to the process of convolving the image with a 2D Gaussian kernel. The Gaussian kernel is the same as the one obtained by solving the heat equation. Hence the action of this kernel on the image is equivalent to the process of isotropic diffusion.

Anisotropic linear diffusion can be generalized from the isotropic linear diffusion by relaxing the requirement for invariance under rotation. This gives us the affine gaussian scale space. Now, this generalized filter lets us perform direction dependent cleaning.

2.2 Non Linear Diffusion

A PDE formulation for adaptive smoothening methods was first given by Perona and Malik. This is a nonlinear diffusion process which makes use of the local properties of the image. In their paper, they made use of scalar diffusivity and not a diffusivity tensor. Hence it is an isotropic process. This can be extended to an anisotropic diffusion process which makes use of a diffusion tensor instead of scalar diffusivity.

2.2.1 Perona Malik Model

This is a model used to avoid blurring and localization problems of linear diffusion filtering. This is an adaptive model. The diffusivity of the model is reduced at those locations where edges are more likely. The likelihood is measured using the quantity $|\nabla u|^2$. Visually, the performance of the PM model is very good. Edges are preserved. This is because edge detection and low pass filtering of noise happen in one single process. This model also has good contrast enhancement properties. However, this model is unstable and illposed.

2.2.2 Cattel-Lion Diffusion

In Perona Malik diffusion scheme, sharp variation of noise is interpreted as edges. It is also unstable and illposed. To avoid this the diffusivity is replaced by a smoothened version. The smoothening of the diffusivity is carried out by convolving the diffusivity tensor with a Gaussian.



FIGURE: Simulation results of the Perona Malik scheme.

2.3 Edge Enhancing Diffusion

In the Perona Malik method, the process acts as a linear diffusion process in the interior of the edges, and hence removing noise. However, it fails to remove noise present at the edges. Therefore we go for Edge Enhancing Diffusion technique. In this technique smoothening of the image is carried out with simultaneous edge enhancement. There is contrast enhancement in anisotropic diffusion filters like in the case of Perona Malik model, provided the regularization parameter is not too large. Here, the two regions with almost constant grey value evolve which are separated by a steep edge. In this technique edge enhancement is caused by rapidly decreasing diffusivities. Intra-region smoothening is preferred over inter-region smoothening between two adjacent regions. The location of the edges remain unchanged over a long period of time. Hence determining the stopping time for cleaning is not an issue. After the process of contrast enhancement is complete, the steepness of edges decreases very slowly until gradient reaches a value where no backward diffusion is possible. After backward diffusion stops, the image quickly turns into a constant image.



FIGURE: MATLAB Simulation results of nonlinear anisotropic on "Lenna" image.

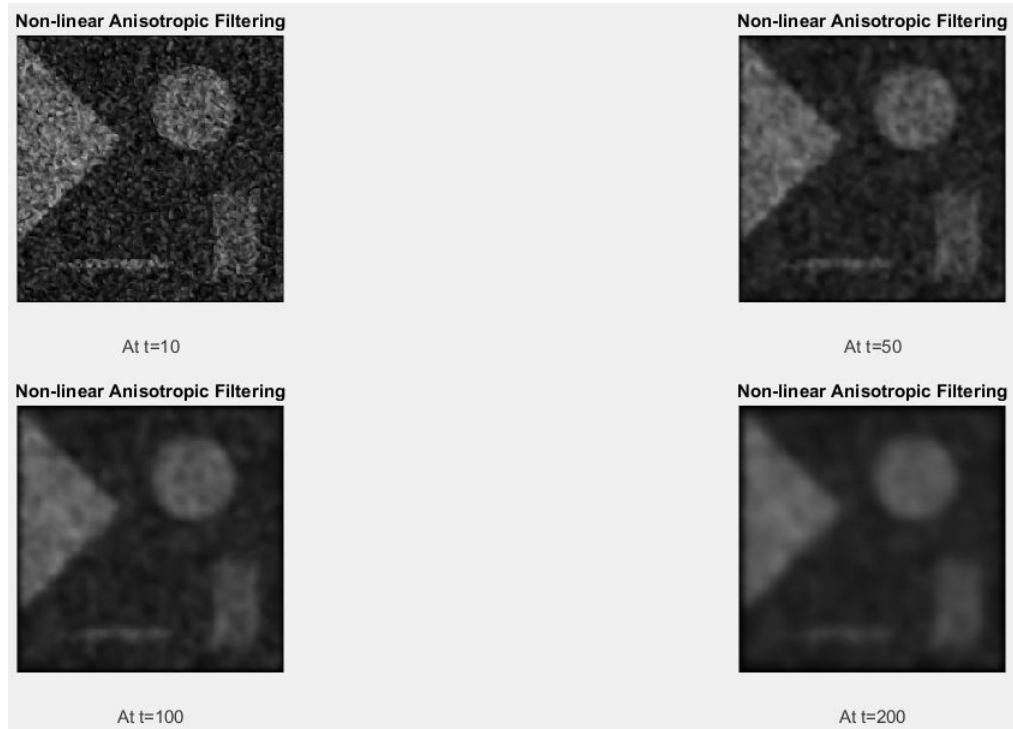


FIGURE: MATLAB Simulation results of nonlinear anisotropic on "synimgn2" image.

2.4 Coherence Enhancing Diffusion

In this technique, structure tensor information is used to design anisotropic scale spaces which enhance the coherence of flow like textures. This technique is very useful in applications like fingerprint analysis where the image contains flow like texture. It is also useful to have a multiscale simplification of the original image by embedding it into a scale-space in order to obtain a subsequently coarser, more global impression of the main flow-like structures. Such a scale-space should take into account the coherence of the structures by smoothing mainly along their preferred orientation instead of perpendicular to it.

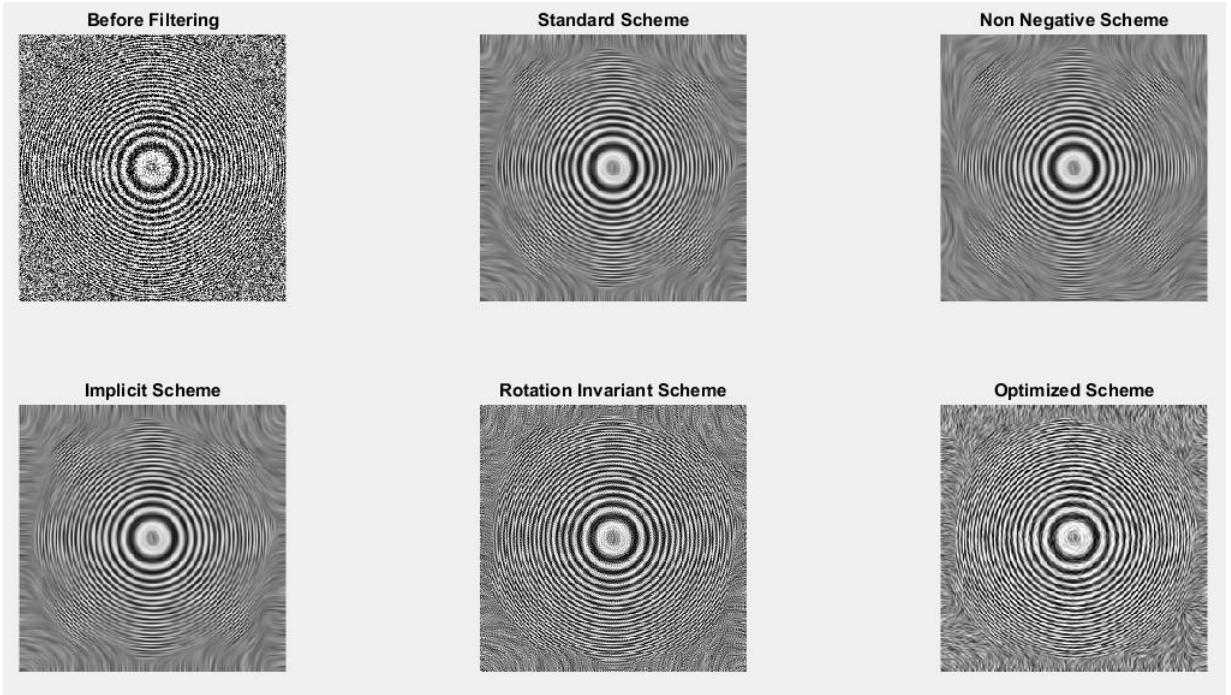


FIGURE: Simulation results of Coherence Enhanced Diffusion for different discretization schemes.

3 Comparison of different filters

In linear diffusion filtering, there is potential to remove all noise but the image becomes completely blurred. Edges are also smoothened along with the noise. Not only do the edges get smoothened but they also get dislocated. Once they are located at coarser scale, they have to be mapped back to their true location. This is hard to do.

In isotropic non linear diffusion like Perona Malik diffusion, the problem of dislocation of edges doesn't occur. However, in this process the diffusivity near the edges is very low. Thus noise near the edges is also preserved.

The PM diffusion is unstable and illposed. It recognizes sharp variations of noise as edges. The PMC method handles this problem by replacing the diffusivity by a smoothened version. The smoothening is carried out by convolving the scalar diffusivity with a Gaussian kernel.

Nonlinear anisotropic filters has advantages of both methods. We get good noise reduction performance as well as stable edge structure. Since smoothening is allowed at the edges, the edges become rounded. The use of diffusion tensors instead of scalar-valued diffusivities puts us in a position to design real anisotropic diffusion processes which may reveal advantages at noisy edges. The fact that these filters are steered by the structure tensor instead of the regularized gradient allows to adapt them to more sophisticated tasks such as the enhancement of coherent flow-like structures. Thus CED is effective in applications like fingerprint scanning.

4 Relationship between parameters σ , λ and T

The parameter σ denotes the standard deviation of the Gaussian kernel. As σ increases the number of amount of averaging increases. Thus in linear filtering, the image becomes more blurred as σ increases. The parameter T denotes the stopping time for smoothening structures of order σ . The stopping time T is related to σ as follows: $T = \frac{\sigma^2}{2}$.

T is the optimal time to carry out the cleaning process. If the cleaning time is less than T , then the image is not cleaned properly. If the cleaning time is greater than T , then the image gets blurred.

λ denotes the contrast parameter in Perona Malik diffusion scheme. It shows up in the expression for scalar diffusivity.

$$g(s^2) = \frac{1}{1 + \frac{s^2}{\lambda^2}}$$

As a result the expression for flux becomes,

$$\Phi(s) = \frac{s}{1 + \frac{s^2}{\lambda^2}}$$

When the gradient is less than λ , the flux is increasing in nature. Thus regions with gradient less than λ are diffused. Thus this region will have low contrast. Similarly, the region where gradient is greater than λ will have high contrast.